January 2026 CSE 220
Online on DFT and FFT

Fourier Magnitude–Phase Swap

Subsections: B1, B2

Total Marks: 10

Total Time: 40 Minutes

September 5, 2026

Every DFT coefficient is complex and therefore has both a magnitude and a phase. The magnitude describes the strength of a spatial frequency; the phase describes how its oscillation is positioned. In this problem you will exchange these parts between two images and observe which part preserves most of the recognizable structure.

You are given the existing sunset and skyline images in `magnitude_phase_swap_template.py`. Complete only the three blocks marked "TODO". Do not modify `transforms.py`, `image_conv.py`, or any other offline file.

Mathematical model. For two real image planes $a[r, c]$ and $b[r, c]$, let

$$A = \mathrm{DFT2}\{a\}, \qquad B = \mathrm{DFT2}\{b\}.$$

For a nonzero complex value Z, its phase-only value is $Z/|Z|$. Values with magnitude no greater than the supplied tolerance ε must be handled without division by zero. Construct

$$S_{AB} = |A| \text{ phase}(B), \qquad S_{BA} = |B| \text{ phase}(A),$$

and inverse-transform both spectra. Thus the first output uses the magnitude of a and phase of b; the second uses the magnitude of b and phase of a. For real images, conjugate symmetry makes both inverse results real apart from small numerical roundoff.

Tasks

1. Complete `unit_phase`. Normalize bins with magnitude greater than `epsilon`; give all remaining bins a safe neutral phase. The function must never divide by zero. [3 Marks]
2. Complete `swap_plane_spectra`. Transform both planes, create S_{AB} and S_{BA}, inverse-transform them, and return their real components. Do not exchange entire complex spectra. [4 Marks]
3. Complete `swap_images`. Process one pair of grayscale planes or three pairs of corresponding RGB planes and return both reconstructed images with their original shape. [3 Marks]

Running and verification. First run the unmodified template and observe the indicated incomplete TODO. After completing all three TODOs, run:

```
python3 magnitude_phase_swap_template.py \
    --image-a images/sunset512.png \
    --image-b images/skyline512.png --engine fft \
    --out-dir outputs/lab_phase_swap_student
```

The runner writes both reconstructed images, a comparison figure, and a report. It verifies the requested magnitude and phase. The report must show `verification: MATCH`; an error above 10^{-9} indicates an incorrect implementation.

Use only NumPy array arithmetic and functions imported by the template; `numpy.fft`, SciPy transforms, and library convolution routines are prohibited. Both grayscale and RGB inputs must work.